



Computer Network

IPv6



The **INTERNET** *of* **THINGS**



During 2008, the number of **things** connected to the Internet exceeded the number of **people** on earth.



2003



2010



2015

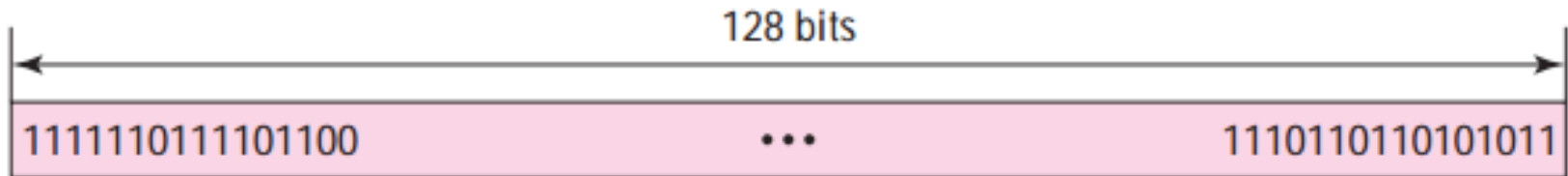


By 2020 there will be **50 billion**.



IPv6 Introduction

- An IPv6 address is 128 bits or 16 bytes long.
- The address length in IPv6 is four times of the length address in IPv4





IPv6 Notation

- Dotted-Decimal : rarely used

221.14.65.11.105.45.170.34.12.234.18.0.14.0.115.255

- Colon Hexadecimal

FDEC : BA98 : 7654 : 3210 : ADBF : BBFF : 2922 : FFFF



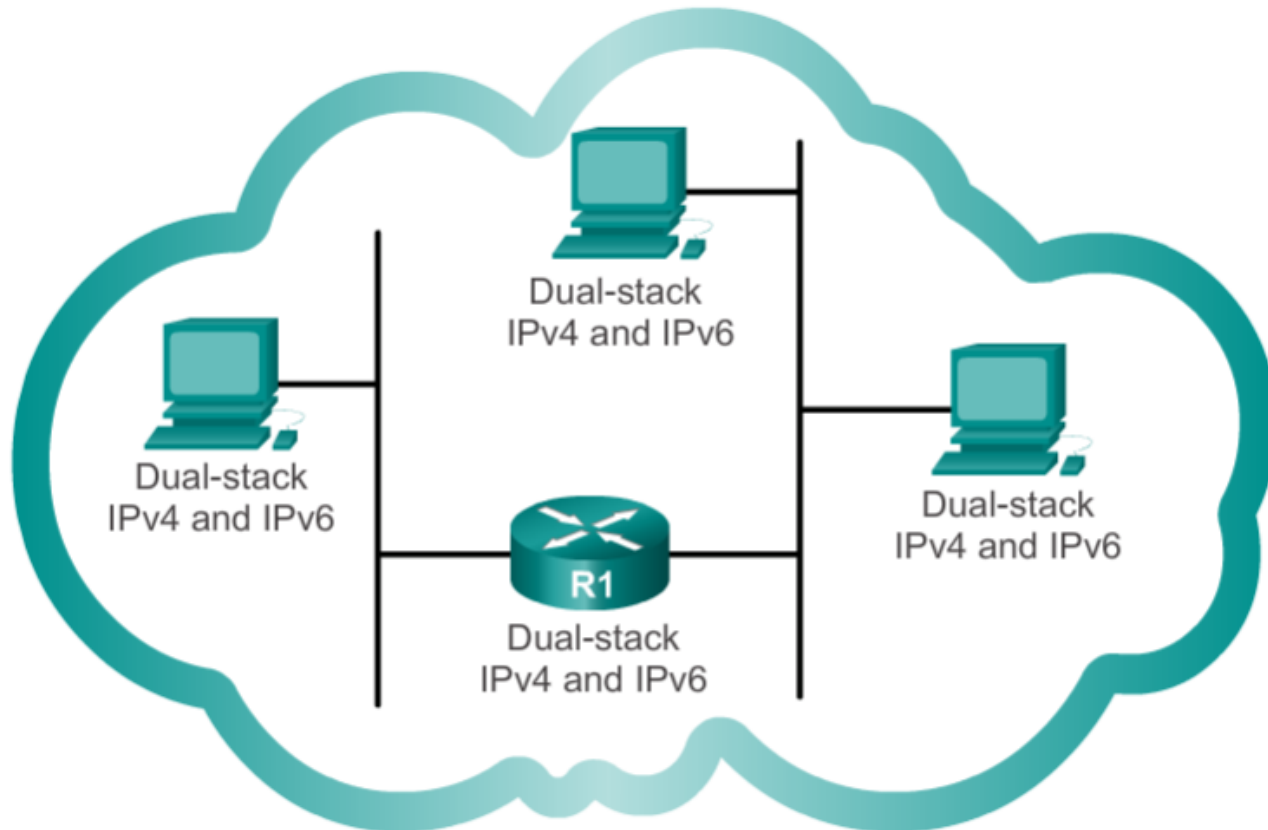
IPv4 and IPv6 Coexistence



- Dual Stack
- Tunneling
- Translation

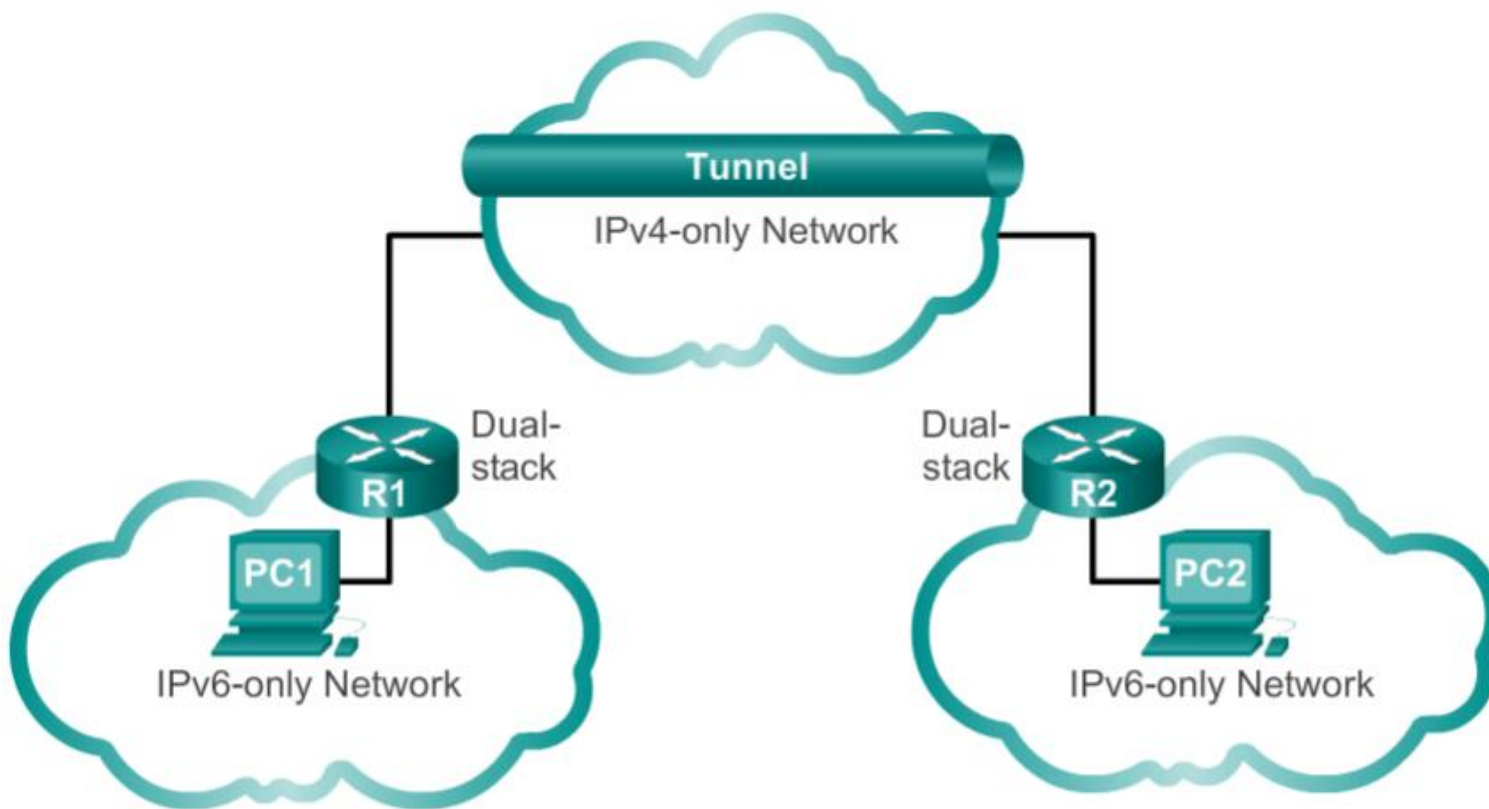


Dual Stack



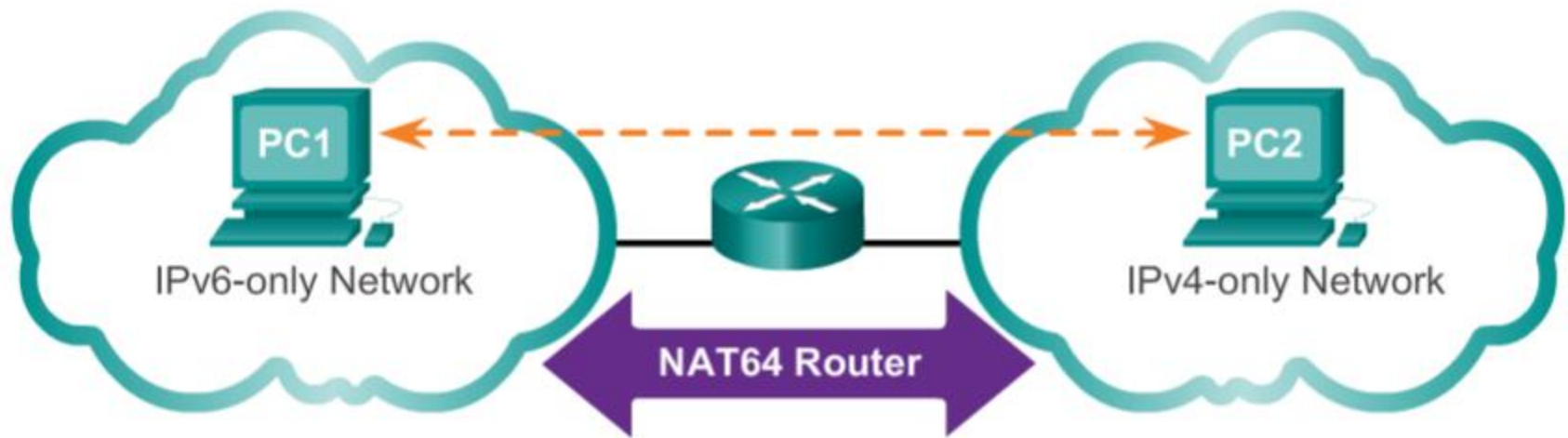


Tunneling





Translation





IPv6 Representation

- Rule 1: any leading 0s (zeros) in any 16-bit section or hextet can be omitted
- Example:
 - 01AB → 1AB
 - 09F0 → 9F0
 - 0A00 → A00
 - 00AB → AB

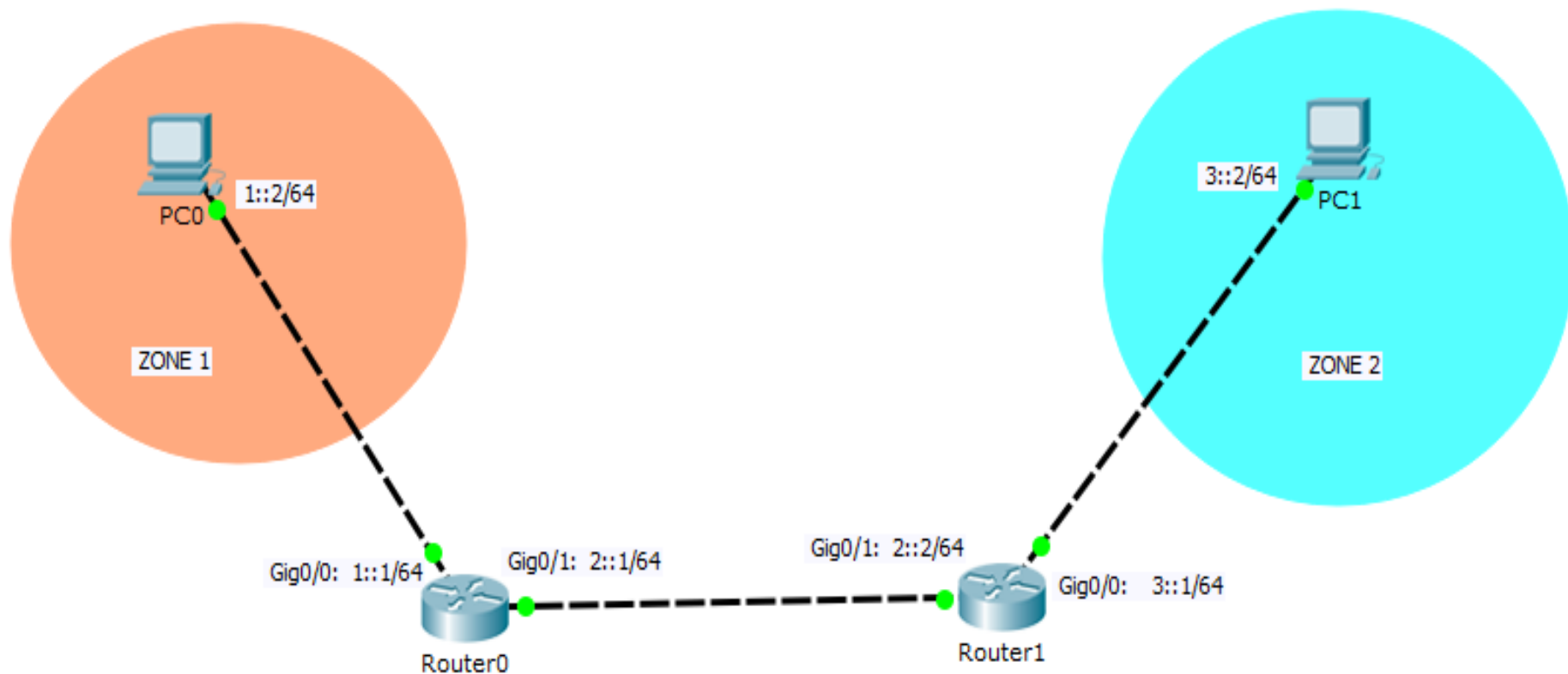


IPv6 Representation

- Rule 2: a double colon (::) can replace any single, contiguous string of one or more 16-bit segments (hexets) consisting of all 0s.
- (::) can only be used once within an address
- Example:
 - 2001:0DB8:0000:0000:ABCD:0000:0000:0100
 - → 2001:0DB8::ABCD:0:0:100
 - OR → 201:0DB8:0:0:ABCD::100



IPv6 Static Routing





IPv6 Static Routing (cnt)



I. Configure IPv6 addresses on Routers

Router 0	Router 1
<pre>Router>enable Router#config terminal Router(config)#ipv6 unicast-routing Router(config)#interface gigabitEthernet0/0 Router(config-if)#ipv6 address 1::1/64 Router(config-if)#no shut Router(config-if)#exit Router(config)#interface gigabitEthernet0/1 Router(config-if)#ipv6 address 2::1/64 Router(config-if)#no shut</pre>	<pre>Router>enable Router#config terminal Router(config)#ipv6 unicast-routing Router(config)#interface gigabitEthernet0/0 Router(config-if)#ipv6 address 3::1/64 Router(config-if)#no shut Router(config-if)#exit Router(config)#interface gigabitEthernet0/1 Router(config-if)#ipv6 address 2::2/64 Router(config-if)#no shut</pre>



IPv6 Static Routing (cnt)

II. Configure IPv6 address on PCs

1. On PC 0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: 1::2 / 64

Link Local Address: FE80::20C:85FF:FE37:EDD5

IPv6 Gateway: 1::1

IPv6 DNS Server:

2. On PC 1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: 3::2 / 64

Link Local Address: FE80::2D0:97FF:FE16:A0A7

IPv6 Gateway: 3::1

IPv6 DNS Server:



IPv6 Static Routing (cnt)



III. Configure IPv6 static route on Routers

Router 0	Router 1
Router>enable Router#config terminal Router(config)#ipv6 route 3::/64 2::2	Router>enable Router#config terminal Router(config)#ipv6 route 1::/64 2::1



IPv6 Static Routing (cnt)



III. Results

PING FROM PC0 to PC1

```
Packet Tracer PC Command Line 1.0
PC>ping 3::2

Pinging 3::2 with 32 bytes of data:

Reply from 3::2: bytes=32 time=1ms TTL=126
Reply from 3::2: bytes=32 time=1ms TTL=126
Reply from 3::2: bytes=32 time=0ms TTL=126
Reply from 3::2: bytes=32 time=0ms TTL=126

Ping statistics for 3::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PING FROM PC1 to PC0

```
Packet Tracer PC Command Line 1.0
PC>ping 1::2

Pinging 1::2 with 32 bytes of data:

Reply from 1::2: bytes=32 time=1ms TTL=126
Reply from 1::2: bytes=32 time=0ms TTL=126
Reply from 1::2: bytes=32 time=0ms TTL=126
Reply from 1::2: bytes=32 time=0ms TTL=126

Ping statistics for 1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```